

ines

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Chapter 1

Design decisions of the ines spec

1.1 Sets

A *set* groups certain entities and gives them common parameters for common constraints.

1.2 Temporal structure

A lot of the preprocessing of the temporal structure is defined by the *solve_pattern*.

The *solve_mode* dictates whether the model uses a single time horizon or a rolling time horizon (single by default). Rolling time horizon means that the same model is used for different horizons and that the input of the next horizon depends on the results of the current horizon. This is not reflected in the model formulation but rather in an algorithm.

The time horizon of every solve pattern is described by a *start_time* and a *duration*. It is sufficient to only give a *duration* value as it starts from the start of the horizon(?) If the solve mode is a rolling horizon, in addition you define the *rolling_horizon* which is also a duration but is smaller than the *duration* of the solve pattern. Each roll will have a length equal to the rolling horizon and will roll as many times as needed to reach the duration of the solve pattern. You can also define a *rolling_jump* to have a gap between each roll. The ines spec currently does not allow for variable roll durations and jumps as the only accepted value is a duration value.

includes_solve_pattern allows for nested setups. For example, you can combine a rolling model hydro model with a rolling dispatch model. When the hydro model includes the dispatch model it means that first the hydro model will roll, then the dispatch model will roll as many times within the rolling horizon of the hydro model (assuming the rolling horizon of the dispatch model is smaller than the rolling horizon of the hydro model).

For rolling and nested patterns to communicate with each other, there are some parameters in the nodes, connections and units to set the method to bind them by various solves (i.e. Benders decomposition). However, the parameters for which this holds is not complete (e.g. a method value 'fix based on upper level').

The *time_resolution* indicates how the data should be sampled. For example, hourly data can be sampled as two hourly data. Different resolutions in different parts of the system are realised with a *set*. The *time_resolution_scope* indicates whether the data has different time resolutions defined by sets or not. The time resolution of the solve pattern then needs to have the smallest possible resolution to be able to cover all time steps in the *timeline* (see later).

period is meant for a larger time scale, e.g. an investment time scale. It is defined by a *start_time* and a *years_represented*. Typically expressed in years, but sometimes there are also quarterly investments that is less than a year. $p_{period}^{years_represented}$ is used to derive the time steps that belong to a period. The used periods also need to be defined in the solve pattern.

The *system* entity has a parameter called the *timeline* and holds all of the allowed datetimes in the model. A lot of the conversion tools expect this timeline to be there. The timeline is a timeseries with each datetime representing a timestep and each value representing the duration which is a float that is to be interpreted as hours (even though one could expect to use a map with datetimes and durations, however, most tools use hours and it would make the conversion tools more difficult to develop).

1.3 Stochastic structure

The stochastic structure is organised as an alternative parameter value, i.e. you can either provide a deterministic value (e.g. *profile_limit_lower*) or a forecast value (e.g. *profile_limit_lower_forecast*). The forecasts can either be independent or branching. The branching forecasts require an additional dimension (i.e. a distinction between analysis time and forecast time?). The *stochastic_scope* method parameter (in the solve pattern) determines which stochastic structure the model uses. *whole_model* implies that the entire model uses the same stochastic structure and every parameter needs to be stochastic if it can be. *single_set* implies that the entire model uses the same stochastic structure but not all parameters need to be stochastic. *multiple_sets* implies that different parts of the model use different stochastic structures.

1.4 Economic representation

The ines spec assumes that the distribution of annuities is flat, i.e. there is no lead time. M_{period} is the milestoneyear of the period; could be start of the simulation for all periods.

$$annuity_{unit,period} = \frac{p_{unit,period}^{interest_rate}}{1 - (1 + p_{unit,period}^{interest_rate})^{-p_{unit}^{lifetime}}}$$

$$discount_factor_{unit,period} = \sum_{y=Y_{period}}^{Y_{period}+p_{unit}^{lifetime}} \frac{1}{1 - (1 + p_{unit,period}^{interest_rate})^{y-M_{period}}}$$

$$\begin{aligned}
\text{salvage factor}_{\text{unit},\text{period}} &= \begin{cases} 0 & Y_{\text{end}} \geq Y_{\text{period}} + p_{\text{unit}}^{\text{lifetime}} \\ \sum_{y=Y_{\text{end}}}^{Y_{\text{period}}+p_{\text{unit}}^{\text{lifetime}}} \frac{1}{(1+p_{\text{unit},\text{period}}^{\text{interest_rate}})^{y-M_{\text{period}}}} & Y_{\text{end}} < Y_{\text{period}} + p_{\text{unit}}^{\text{lifetime}} \end{cases} \\
p_{\text{unit},\text{period}}^{\text{economic_investment}} &= \text{annuity}_{\text{unit},\text{period}} \cdot (\text{discount factor}_{\text{unit},\text{period}} - \text{salvage factor}_{\text{unit},\text{period}}) \\
p_{\text{unit},\text{period}}^{\text{economic_fixed}} &= \begin{cases} \frac{p_{\text{period}}^{\text{years_represented}}}{1 - (1+p_{\text{unit},\text{period}}^{\text{interest_rate}})^{Y_{\text{period}}-M_{\text{period}}}} & p_{\text{period}}^{\text{years_represented}} < 1 \\ \sum_{y=Y_{\text{period}}}^{Y_{\text{period}}+p_{\text{period}}^{\text{years_represented}}} \frac{1}{1 - (1+p_{\text{unit},\text{period}}^{\text{interest_rate}})^{y-M_{\text{period}}}} & p_{\text{period}}^{\text{years_represented}} \geq 1 \end{cases} \\
p_{\text{unit},\text{period}}^{\text{economic_other}} &= \begin{cases} \frac{1}{1 - (1+p_{\text{unit},\text{period}}^{\text{interest_rate}})^{Y_{\text{period}}-M_{\text{period}}}} & p_{\text{period}}^{\text{years_represented}} < 1 \\ \sum_{y=Y_{\text{period}}}^{Y_{\text{period}}+p_{\text{period}}^{\text{years_represented}}} \frac{1}{1 - (1+p_{\text{unit},\text{period}}^{\text{interest_rate}})^{y-M_{\text{period}}}} & p_{\text{period}}^{\text{years_represented}} \geq 1 \end{cases}
\end{aligned}$$

1.5 units, nodes, links and flows

Units and links operate between nodes. A node has a *node_type* which indicates how the node is balanced. *balance* implies that all incoming and outgoing flows need to equal each other. *commodity* implies the use of a commodity to deal with the imbalance of flows in the node. A typical configuration is a 'source' node with a *commodity* node type connected to a unit (through a *node_to_unit* entity) and the unit connected to a 'demand' node with a *balanced* node type (through a *unit_to_node* entity). The 'demand' node would then also have a parameter *demand* which denotes the demand of the node in every time step.

The efficiency etc. is defined at the unit level, whereas the capacity is defined at the flow level (although the flow can also define an efficiency that is more flexible, similar to SpineOpt). The capacity needs to be defined for every flow. The efficiency needs to be defined for every unit. The capacity is per unit as it is multiplied with the number of existing units, which means that both the capacity and the number of existing units always need to be specified. The direction of the flow is handled by *node_link_node*, on one hand in the transfer method in *link* and on the other hand with multiplier from *constraint_flow*. To deal with multiple profiles for a unit in a multiyear setup, you can define a unit with the same parameters but different profiles (though in some setups you would require a map with a period and a time, which is currently not supported).

unit_conversion_method provides a few conversion methods including *partial_load_efficiency* and *piecewise_linear* as depicted in the figure below.

To define a cyclic condition for storage in a node, you set the *storage_binding_method* to leap over a period or solve. You can also fix the initial state of the storage with the *storage_fix_state_method*.

Across the entities there are some scaling parameters (methods) that can be used to, e.g., scale the flow of a partial solve to a year.

Some entities also have methods for investments. When it is cumulative, it describes over multiple periods what is the total allowed capacity. Which can also be set for a set.

To set up a user constraint you define a *constraint* with the *sense* and *constant*. You can then link a *unit_to_node* entity to this constraint through the *constraint_flow_coefficient*.

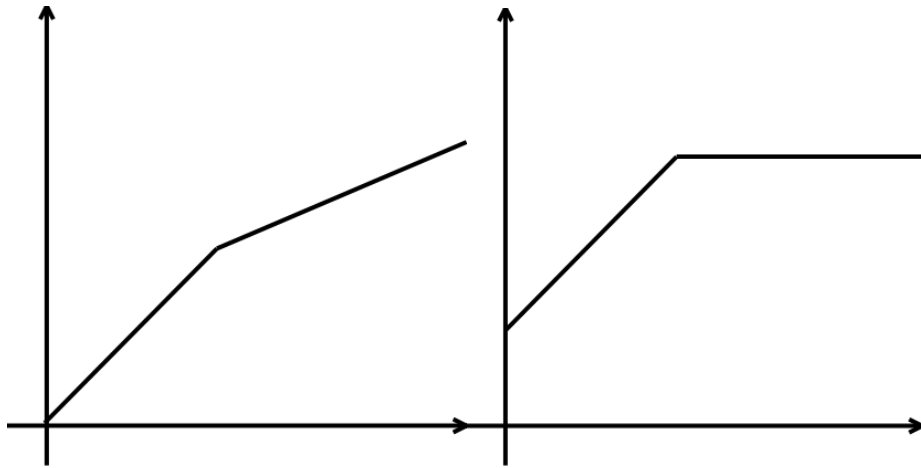


Figure 1.1: Visual depiction of *unit_conversion_method*: piece wise linear (left) and partial load efficiency (right)

Chapter 2

Proof of concept for the certification process

As the ines spec describes an interoperable data format, it is particularly interesting to define a certification process in that format such that it can be used to certify different energy system models for the correct implementation of certain features.

Here, we only consider a proof of concept to show the potential of certification with the ines spec. Once a consortium has been formed around the ines spec, it makes sense to continue the development of the certification process. In the following, we cherry pick some setups that show different aspects of energy system models. Some suggestions for expansions on these setups can be found in [appendix C](#).

While it is possible to describe different features in the ines spec, there is not actually a specific model associated to it. Therefore, if we want to obtain the exact solution for the certification process, we'll have to solve the system manually. In the following, not only do we provide a description of the setup but also provide the specific equations to solve that setup.

2.1 network = mesh

Triangular network with the two supply units and demand unit. The cheapest unit should be chosen again.

minimise:

$$z = p_{cheap_source}^{other_operational_costs} \cdot v_{cheap_source}^{flow} + p_{expensive_source}^{other_operational_costs} \cdot v_{expensive_source}^{flow}$$

subject to:

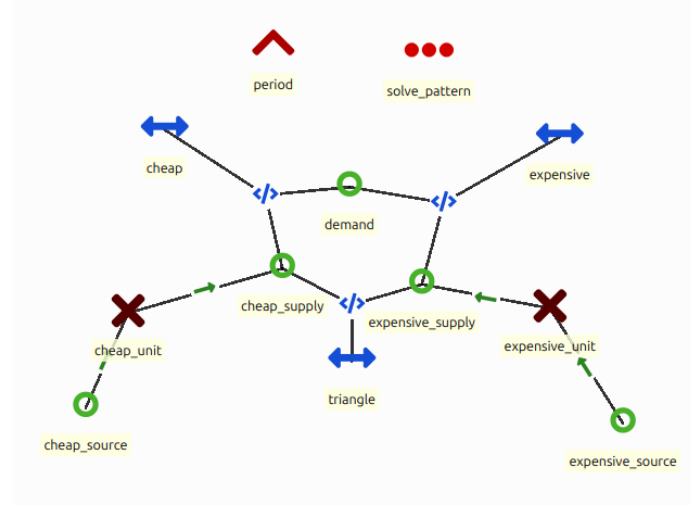


Figure 2.1: Network = mesh

$$\begin{aligned}
 p_{demand}^{flow_profile} &= v_{cheap_link,out}^{flow} + v_{expensive_link,out}^{flow} \\
 v_{cheap_link,out}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in}^{flow} \\
 v_{expensive_link,out}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in}^{flow} \\
 v_{triangle,out}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in}^{flow} \\
 v_{cheap_link,in}^{flow} &= v_{cheap_supply}^{flow} + v_{triangle,out}^{flow} \\
 v_{expensive_supply}^{flow} &= v_{expensive_link,in}^{flow} + v_{triangle,in}^{flow} \\
 v_{expensive_supply}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source}^{flow} \\
 v_{cheap_supply}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source}^{flow}
 \end{aligned}$$

bound to:

$$\begin{aligned}
0 &\leq v_{cheap_link,in}^{flow} \\
0 &\leq v_{cheap_link,out}^{flow} && \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\
0 &\leq v_{expensive_link,in}^{flow} \\
0 &\leq v_{expensive_link,out}^{flow} && \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\
0 &\leq v_{triangle,in}^{flow} \\
0 &\leq v_{triangle,out}^{flow} && \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
0 &\leq v_{expensive_supply}^{flow} && \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\
0 &\leq v_{expensive_source}^{flow} \\
0 &\leq v_{cheap_supply}^{flow} && \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \\
0 &\leq v_{cheap_source}^{flow}
\end{aligned}$$

2.2 resolution = fixed

Same setup but with a time series for all units with the same resolution.

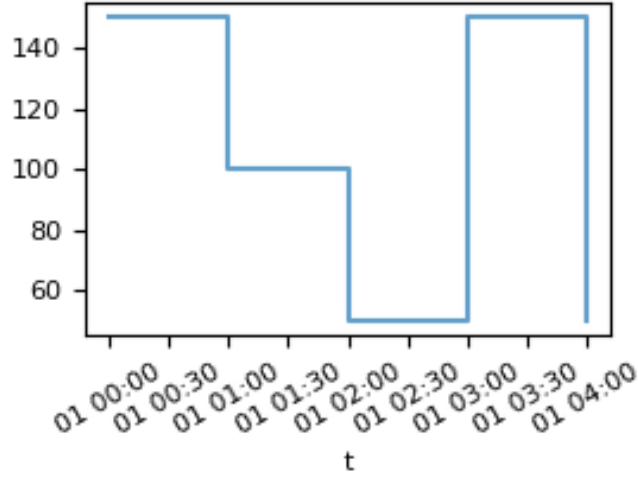


Figure 2.2: resolution = fixed

minimise:

$$z = \sum_t p_{cheap_source,t}^{other_operational_costs} \cdot v_{cheap_source,t}^{flow} + p_{expensive_source,t}^{other_operational_costs} \cdot v_{expensive_source,t}^{flow}$$

subject to:

$$\begin{aligned}
p_{demand,t}^{flow_profile} &= v_{cheap_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow} \\
v_{cheap_link,out,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow} \\
v_{expensive_link,out,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow} \\
v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow} \\
v_{cheap_link,in,t}^{flow} &= v_{cheap_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\
v_{expensive_supply,t}^{flow} &= v_{expensive_link,in,t}^{flow} + v_{triangle,in,t}^{flow} \\
v_{expensive_supply,t}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t}^{flow} \\
v_{cheap_supply,t}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t}^{flow}
\end{aligned}$$

bound to:

$$\begin{aligned}
0 &\leq v_{cheap_link,in,t}^{flow} \\
0 &\leq v_{cheap_link,out,t}^{flow} && \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\
0 &\leq v_{expensive_link,in,t}^{flow} \\
0 &\leq v_{expensive_link,out,t}^{flow} && \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\
0 &\leq v_{triangle,in,t}^{flow} \\
0 &\leq v_{triangle,out,t}^{flow} && \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
0 &\leq v_{expensive_supply,t}^{flow} && \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\
0 &\leq v_{expensive_source,t}^{flow} \\
0 &\leq v_{cheap_supply,t}^{flow} && \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \\
0 &\leq v_{cheap_source,t}^{flow}
\end{aligned}$$

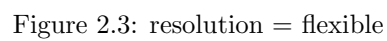
2.3 resolution = flexible

Same setup but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).

minimise:

$$z = \sum_{t \in t_{cheap}} p_{cheap_source,t}^{other_operational_costs} \cdot v_{cheap_source,t}^{flow} + \sum_t p_{expensive_source,t}^{other_operational_costs} \cdot v_{expensive_source,t}^{flow}$$

subject to:



bound to:

$$\begin{aligned}
0 &\leq v_{cheap_link,in,t}^{flow} \\
0 &\leq v_{cheap_link,out,t}^{flow} && \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\
0 &\leq v_{expensive_link,in,t}^{flow} \\
0 &\leq v_{expensive_link,out,t}^{flow} && \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\
0 &\leq v_{triangle,in,t}^{flow} \\
0 &\leq v_{triangle,out,t}^{flow} && \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
0 &\leq v_{expensive_supply,t}^{flow} && \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\
0 &\leq v_{expensive_source,t}^{flow} \\
0 &\leq v_{cheap_supply,t}^{flow} && \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \\
0 &\leq v_{cheap_source,t}^{flow}
\end{aligned}$$

2.4 stochastic = flexible

Meshed network, fixed resolution, expensive and cheap unit. the units have different stochastics.

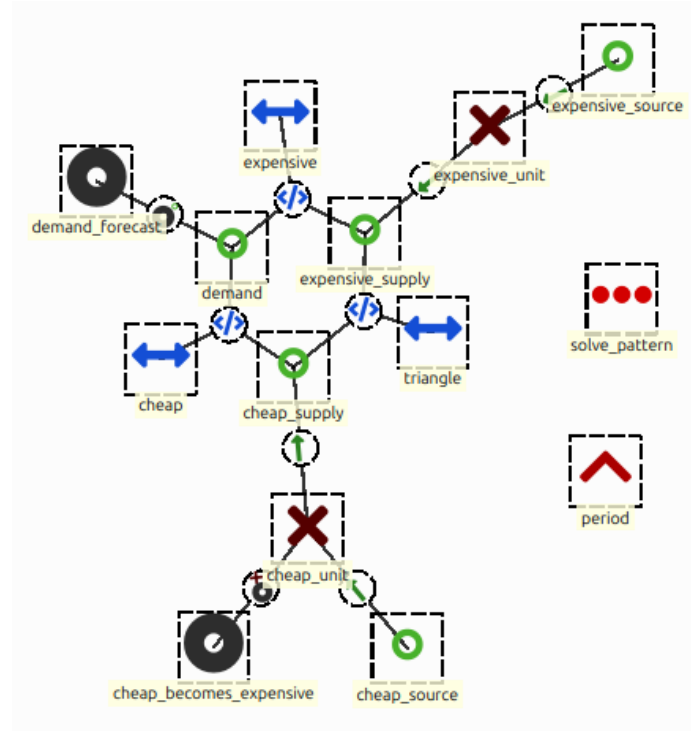


Figure 2.4: stochastic = flexible

minimise:

$$z = \sum_t \sum_s p_{cheap_source,t,s}^{other_operational_costs} \cdot p_{cheap_source,s}^{weight} \cdot v_{cheap_source,t,s}^{flow} + p_{expensive_source,t,s}^{other_operational_costs} \cdot v_{expensive_source,t,s}^{flow}$$

subject to:

$$\begin{aligned} p_{demand,t,s}^{flow_profile} &= v_{cheap_link,out,t,s}^{flow} + v_{expensive_link,out,t,s}^{flow} \\ v_{cheap_link,out,t,s}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t,s}^{flow} \\ v_{expensive_link,out,t,s}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t,s}^{flow} \\ v_{triangle,out,t,s}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t,s}^{flow} \\ v_{cheap_link,in,t,s}^{flow} &= v_{cheap_supply,t,s}^{flow} + v_{triangle,out,t,s}^{flow} \\ v_{expensive_supply,t,s}^{flow} &= v_{expensive_link,in,t,s}^{flow} + v_{triangle,in,t,s}^{flow} \\ v_{expensive_supply,t,s}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t,s}^{flow} \\ v_{cheap_supply,t,s}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t,s}^{flow} \end{aligned}$$

bound to:

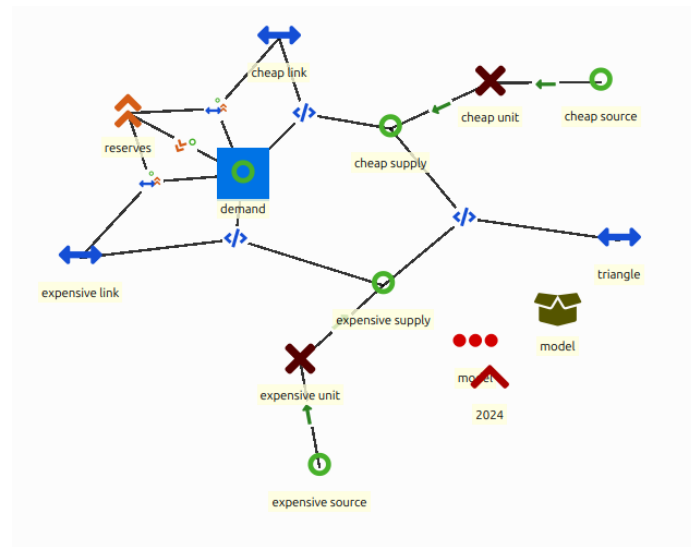
$$\begin{aligned} 0 &\leq v_{cheap_link,in,t,s}^{flow} \\ 0 &\leq v_{cheap_link,out,t,s}^{flow} \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\ 0 &\leq v_{expensive_link,in,t,s}^{flow} \\ 0 &\leq v_{expensive_link,out,t,s}^{flow} \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\ 0 &\leq v_{triangle,in,t,s}^{flow} \\ 0 &\leq v_{triangle,out,t,s}^{flow} \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\ 0 &\leq v_{expensive_supply,t,s}^{flow} \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\ 0 &\leq v_{expensive_source,t,s}^{flow} \\ 0 &\leq v_{cheap_supply,t,s}^{flow} \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \\ 0 &\leq v_{cheap_source,t,s}^{flow} \end{aligned}$$

2.5 reserves

Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.

minimise:

$$z = \sum_t p_{cheap_source,t}^{other_operational_costs} \cdot v_{cheap_source,t}^{flow} + p_{expensive_source,t}^{other_operational_costs} \cdot v_{expensive_source,t}^{flow} + p_{demand}^{penalty} \cdot v_{demand,t}^{slack}$$



subject to:

$$p_{demand,t}^{reserve} = v_{cheap_link,out,t}^{flow,reserve} + v_{expensive_link,out,t}^{flow,reserve} + v_{demand,t}^{slack}$$

$$p_{demand,t}^{flow_profile} = v_{cheap_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow} \\ v_{expensive_link,out,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow} \\ v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow,reserve} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow,reserve} \\ v_{expensive_link,out,t}^{flow,reserve} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow,reserve} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow} + v_{cheap_link,out,t}^{flow,reserve} &\leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_links} \\ v_{expensive_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow,reserve} &\leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_links} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,in,t}^{flow} &= v_{cheap_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\ v_{expensive_supply,t}^{flow} &= v_{expensive_link,in,t}^{flow} + v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{cheap_supply,t}^{flow,reserve} &= v_{cheap_link,in,t}^{flow,reserve} \\ v_{expensive_supply,t}^{flow,reserve} &= v_{expensive_link,in,t}^{flow,reserve} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t}^{flow} \\ v_{cheap_supply,t}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} + v_{expensive_supply,t}^{flow,reserve} &\leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\ v_{cheap_supply,t}^{flow} + v_{expensive_supply,t}^{flow,reserve} &\leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \end{aligned}$$

bound to:

$$\begin{aligned}
0 &\leq v_{demand,t}^{slack} \\
0 &\leq v_{cheap_link,out,t}^{flow,reserve} \leq p_{cheap_link}^{max_reserve} \cdot p_{cheap_link}^{existing_links} \\
0 &\leq v_{expensive_link,out,t}^{flow,reserve} \leq p_{expensive_link}^{max_reserve} \cdot p_{expensive_link}^{existing_links} \\
0 &\leq v_{cheap_supply,t}^{flow,reserve} \leq p_{cheap_unit}^{max_reserve} \cdot p_{cheap_unit}^{existing_units} \\
0 &\leq v_{expensive_supply,t}^{flow,reserve} \leq p_{expensive_unit}^{max_reserve} \cdot p_{expensive_unit}^{existing_units} \\
\\
0 &\leq v_{cheap_link,out,t}^{flow} \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_links} \\
0 &\leq v_{expensive_link,out,t}^{flow} \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_links} \\
0 &\leq v_{triangle,out,t}^{flow} \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
\\
0 &\leq v_{expensive_supply,t}^{flow} \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\
0 &\leq v_{cheap_supply,t}^{flow} \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units}
\end{aligned}$$

2.6 Unit commitment

Meshed network, fixed resolution, expensive and cheap unit. Both units have part load efficiency, MUT/MDT and startup cost. The startup cost of the cheap unit is large.

minimise:

$$z = \sum_t p_{cheap_unit}^{startup_cost} \cdot v_{cheap_unit,t}^{on-up} + p_{cheap_source,t}^{other_operational_costs} \cdot v_{cheap_source,t}^{flow} + p_{expensive_unit}^{startup_cost} \cdot v_{expensive_unit,t}^{on-up} + p_{expensive_source,t}^{other_operational_costs} \cdot v_{expensive_source,t}^{flow}$$

subject to:

$$p_{demand,t}^{flow_profile} = v_{cheap_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow} \\ v_{expensive_link,out,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow} \\ v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,in,t}^{flow} &= v_{cheap_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\ v_{expensive_supply,t}^{flow} &= v_{expensive_link,in,t}^{flow} + v_{triangle,in,t}^{flow} \end{aligned}$$

$$v_{expensive_supply,t}^{flow} = p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t}^{flow} + \left(1 - \frac{p_{expensive_unit}^{efficiency}}{p_{expensive_unit}^{efficiency_min}}\right) \cdot p_{expensive_unit}^{capacity_min} \cdot p_{expensive_unit}^{existing_units} \cdot$$

$$v_{cheap_supply,t}^{flow} = p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t}^{flow} + \left(1 - \frac{p_{cheap_unit}^{efficiency}}{p_{cheap_unit}^{efficiency_min}}\right) \cdot p_{cheap_unit}^{capacity_min} \cdot p_{cheap_unit}^{existing_units} \cdot v_{cheap_unit}^{on}$$

$$\begin{aligned} v_{cheap_unit,t}^{on} &= v_{cheap_unit,t-1}^{on} + v_{cheap_unit,t}^{on_up} - v_{cheap_unit,t}^{on_down} \\ v_{expensive_unit,t}^{on} &= v_{expensive_unit,t-1}^{on} + v_{expensive_unit,t}^{on_up} - v_{expensive_unit,t}^{on_down} \end{aligned}$$

$$v_{cheap_unit,t}^{on_up} \leq 1 - v_{cheap_unit,t}^{on} - \sum_{t'=1}^{MDT-1} v_{cheap_unit,t-t'}^{on_down}$$

$$v_{cheap_unit,t}^{on_down} \leq v_{cheap_unit,t}^{on} - \sum_{t'=1}^{MUT-1} v_{cheap_unit,t-t'}^{on_up}$$

$$v_{expensive_unit,t}^{on_up} \leq 1 - v_{expensive_unit,t}^{on} - \sum_{t'=1}^{MDT-1} v_{expensive_unit,t-t'}^{on_down}$$

$$v_{expensive_unit,t}^{on_down} \leq v_{expensive_unit,t}^{on} - \sum_{t'=1}^{MUT-1} v_{expensive_unit,t-t'}^{on_up}$$

bound to:

$$\begin{aligned}
0 &\leq v_{cheap_link,in,t}^{flow} \\
0 &\leq v_{cheap_link,out,t}^{flow} && \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\
0 &\leq v_{expensive_link,in,t}^{flow} \\
0 &\leq v_{expensive_link,out,t}^{flow} && \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\
0 &\leq v_{triangle,in,t}^{flow} \\
0 &\leq v_{triangle,out,t}^{flow} && \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
p_{expensive_unit}^{capacity_min} \cdot p_{expensive_unit}^{existing_units} \cdot v_{expensive_unit,t}^{on} &\leq v_{expensive_supply,t}^{flow} && \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \cdot v_{expensive_unit,t}^{on} \\
0 &\leq v_{expensive_source,t}^{flow} \\
p_{cheap_unit}^{capacity_min} \cdot p_{cheap_unit}^{existing_units} \cdot v_{cheap_unit,t}^{on} &\leq v_{cheap_supply,t}^{flow} && \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \cdot v_{cheap_unit,t}^{on} \\
0 &\leq v_{cheap_source,t}^{flow} \\
0 &\leq v_{cheap_unit,t}^{on} \\
0 &\leq v_{cheap_unit,t}^{on_up} \\
0 &\leq v_{cheap_unit,t}^{on_down} \\
0 &\leq v_{expensive_unit,t}^{on} \\
0 &\leq v_{expensive_unit,t}^{on_up} \\
0 &\leq v_{expensive_unit,t}^{on_down}
\end{aligned}$$

2.7 investments

Meshed network, fixed resolution, expensive and cheap unit. Expensive unit is already invested in. The investment cost is to be interpreted as the cost per unit, as opposed to the cost per MW(h).

minimise:

$$z = p_{cheap_unit}^{investment_cost} \cdot v_{cheap_unit}^{investment} + p_{expensive_unit}^{investment_cost} \cdot (v_{expensive_unit}^{investment} - p_{expensive_unit}^{existing_units}) + \sum_t p_{cheap_source,t}^{other_operational_cost}$$

subject to:

$$p_{demand,t}^{flow_profile} = v_{cheap_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow} \\ v_{expensive_link,out,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow} \\ v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,in,t}^{flow} &= v_{cheap_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\ v_{expensive_supply,t}^{flow} &= v_{expensive_link,in,t}^{flow} + v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t}^{flow} \\ v_{cheap_supply,t}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} &\leq p_{expensive_unit}^{capacity} \cdot v_{expensive_unit}^{investment} \\ v_{cheap_supply,t}^{flow} &\leq p_{cheap_unit}^{capacity} \cdot v_{cheap_unit}^{investment} \end{aligned}$$

bound to:

$$\begin{aligned} 0 &\leq v_{cheap_link,in,t}^{flow} \\ 0 &\leq v_{cheap_link,out,t}^{flow} \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\ 0 &\leq v_{expensive_link,in,t}^{flow} \\ 0 &\leq v_{expensive_link,out,t}^{flow} \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\ 0 &\leq v_{triangle,in,t}^{flow} \\ 0 &\leq v_{triangle,out,t}^{flow} \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\ 0 &\leq v_{expensive_supply,t}^{flow} \\ 0 &\leq v_{expensive_source,t}^{flow} \\ 0 &\leq v_{cheap_supply,t}^{flow} \\ 0 &\leq v_{cheap_source,t}^{flow} \\ 0 &\leq v_{cheap_unit}^{investment} \\ p_{expensive_unit}^{existing_units} &\leq v_{expensive_unit}^{investment} \end{aligned}$$

2.8 storage

Meshed network, fixed resolution, inflexible cheap unit, storage. Representative periods with intra and inter storage. Inter is between days and intra is within

a day. The delta formulation mostly means the addition of inter constraints where the delta of the day is added every day.

minimise:

$$z = \sum_r W_r \sum_t p_{cheap_source,r,t}^{other_operational_costs} \cdot v_{cheap_source,r,t}^{flow} + p_{expensive_source,r,t}^{other_operational_costs} \cdot v_{expensive_source,r,t}^{flow}$$

subject to:

$$\begin{aligned} p_{demand,r,t}^{flow_profile} &= v_{cheap_link,out,r,t}^{flow} + v_{expensive_link,out,r,t}^{flow} - v_{storage,r,t}^{charge} + p_{storage}^{efficiency,discharge} \cdot v_{storage,r,t}^{discharge} \\ v_{cheap_link,out,r,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,r,t}^{flow} \\ v_{expensive_link,out,r,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,r,t}^{flow} \\ v_{triangle,out,r,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,r,t}^{flow} \\ v_{cheap_link,in,r,t}^{flow} &= v_{cheap_supply,r,t}^{flow} + v_{triangle,out,r,t}^{flow} \\ v_{expensive_supply,r,t}^{flow} &= v_{expensive_link,in,r,t}^{flow} + v_{triangle,in,r,t}^{flow} \\ v_{expensive_supply,r,t}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,r,t}^{flow} \\ v_{cheap_supply,r,t}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,r,t}^{flow} \\ v_{storage,r,t}^{state,intra} &= p_{storage}^{efficiency} \cdot v_{storage,r,t-1}^{state,intra} + p_{storage}^{efficiency,charge} \cdot v_{storage,r,t}^{charge} - v_{storage,r,t}^{discharge} \\ v_{storage,d}^{state,inter} &= v_{storage,d-1}^{state,inter} + \sum_r W_{d,r} \cdot (v_{storage,r,0}^{state,intra} - v_{storage,r,T}^{state,intra}) \\ v_{storage,0}^{state,inter} &= v_{storage,D}^{state,inter} \end{aligned}$$

bound to:

$$\begin{aligned}
0 \leq v_{cheap_link,out,r,t}^{flow} & \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\
0 \leq v_{expensive_link,out,r,t}^{flow} & \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\
0 \leq v_{triangle,out,r,t}^{flow} & \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\
\\
0 \leq v_{expensive_supply,r,t}^{flow} & \leq p_{expensive_unit}^{capacity} \cdot p_{expensive_unit}^{existing_units} \\
0 \leq v_{cheap_supply,r,t}^{flow} & \leq p_{cheap_unit}^{capacity} \cdot p_{cheap_unit}^{existing_units} \\
\\
0 \leq v_{storage,r,t}^{charge} & \leq p_{storage}^{capacity_charge} \cdot p_{storage}^{existing_units} \\
0 \leq v_{storage,r,t}^{discharge} & \leq p_{storage}^{capacity_discharge} \cdot p_{storage}^{existing_units} \\
0 \leq v_{storage,r,t}^{state,intra} & \leq p_{storage}^{capacity} \cdot p_{storage}^{existing_units} \\
0 \leq v_{storage,d}^{state,inter} & \leq p_{storage}^{capacity} \cdot p_{storage}^{existing_units}
\end{aligned}$$

2.9 rolling horizon

Same setup as investments but there are two investment periods and they are separated.

There are two windows, each uses the same formulation. In the first window, there is already a small investment in expensive units and the model needs to invest in cheaper units. The second window has a higher demand requiring additional investments but there is also a shift in commodity prices, yet there are already the investments from the first window.

minimise:

$$z = p_{cheap_unit}^{investment_cost} \cdot (v_{cheap_unit}^{investment} - p_{cheap_unit}^{existing_units}) + p_{expensive_unit}^{investment_cost} \cdot (v_{expensive_unit}^{investment} - p_{expensive_unit}^{existing_units}) + \sum_t p_t$$

subject to:

$$p_{demand,t}^{flow_profile} = v_{cheap_link,out,t}^{flow} + v_{expensive_link,out,t}^{flow}$$

$$\begin{aligned} v_{cheap_link,out,t}^{flow} &= p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t}^{flow} \\ v_{expensive_link,out,t}^{flow} &= p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t}^{flow} \\ v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{cheap_link,in,t}^{flow} &= v_{cheap_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\ v_{expensive_supply,t}^{flow} &= v_{expensive_link,in,t}^{flow} + v_{triangle,in,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} &= p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t}^{flow} \\ v_{cheap_supply,t}^{flow} &= p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t}^{flow} \end{aligned}$$

$$\begin{aligned} v_{expensive_supply,t}^{flow} &\leq p_{expensive_unit}^{capacity} \cdot v_{expensive_unit}^{investment} \\ v_{cheap_supply,t}^{flow} &\leq p_{cheap_unit}^{capacity} \cdot v_{cheap_unit}^{investment} \end{aligned}$$

bound to:

$$\begin{aligned} 0 &\leq v_{cheap_link,in,t}^{flow} \\ 0 &\leq v_{cheap_link,out,t}^{flow} \leq p_{cheap_link}^{capacity} \cdot p_{cheap_link}^{existing_units} \\ 0 &\leq v_{expensive_link,in,t}^{flow} \\ 0 &\leq v_{expensive_link,out,t}^{flow} \leq p_{expensive_link}^{capacity} \cdot p_{expensive_link}^{existing_units} \\ 0 &\leq v_{triangle,in,t}^{flow} \\ 0 &\leq v_{triangle,out,t}^{flow} \leq p_{triangle}^{capacity} \cdot p_{triangle}^{existing_units} \\ 0 &\leq v_{expensive_supply,t}^{flow} \\ 0 &\leq v_{expensive_source,t}^{flow} \\ 0 &\leq v_{cheap_supply,t}^{flow} \\ 0 &\leq v_{cheap_source,t}^{flow} \\ p_{cheap_unit}^{existing_units} &\leq v_{cheap_unit}^{investment} \\ p_{expensive_unit}^{existing_units} &\leq v_{expensive_unit}^{investment} \end{aligned}$$

2.10 combined system

A combination of all the previous setups. Sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries,

onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. There are stochastics for the renewables and import prices.

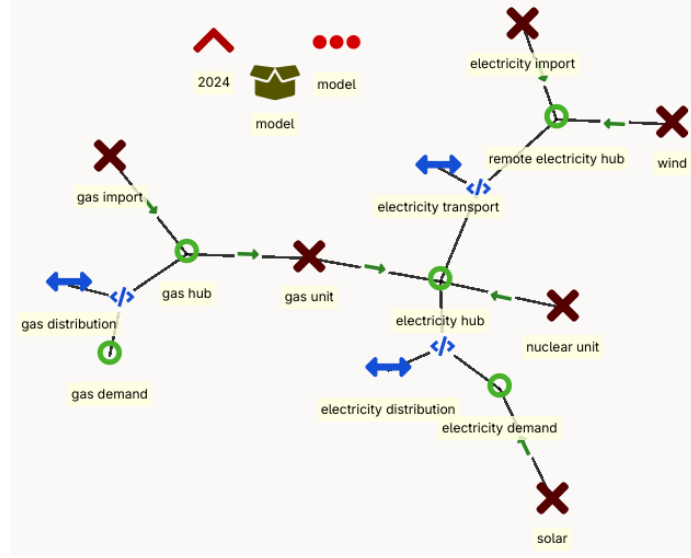


Figure 2.6: combined system

minimise:

$$\begin{aligned}
 z = & \sum_r W_r \sum_s \sum_{t \in t_{gas}} p_{gas_import,r,s,t}^{other_operational_costs} \cdot v_{gas_import,r,s,t}^{flow} + \sum_t p_{electricity_import,r,s,t}^{other_operational_costs} \cdot v_{electricity_import,r,s,t}^{flow} \\
 & + \sum_r W_r \sum_s \sum_t p_{nuclear_unit}^{startup_cost} \cdot v_{nuclear_unit,r,s,t}^{on_up.} + p_{nuclear_unit}^{other_operational_costs} \cdot v_{nuclear_unit,r,s,t}^{flow} \\
 & + p_{gas_unit}^{investment_cost} \cdot (v_{gas_unit}^{investment} - p_{gas_unit}^{existing_units}) + p_{nuclear_unit}^{investment_cost} \cdot (v_{nuclear_unit}^{investment} - p_{nuclear_unit}^{existing_units}) \\
 & + p_{wind_unit}^{investment_cost} \cdot (v_{wind_unit}^{investment} - p_{wind_unit}^{existing_units}) + p_{solar_unit}^{investment_cost} \cdot (v_{solar_unit}^{investment} - p_{solar_unit}^{existing_units}) + p_{battery}^{investment_cost} \cdot (v_{battery_unit}^{investment} - p_{battery_unit}^{existing_units})
 \end{aligned}$$

subject to:

$$\begin{aligned}
p_{electricity_demand,r,t}^{flow_profile} &= v_{electricity_distribution,out,r,s,t}^{flow} + v_{solar_unit,out,r,s,t}^{flow} \\
p_{gas_demand,r,t}^{flow_profile} &= v_{gas_distribution,out,r,s,t}^{flow} \\
v_{gas_import,r,s,t}^{flow} &= v_{gas_distribution,in,r,s,t}^{flow} + v_{gas_unit,in,r,s,t}^{flow} \\
v_{electricity_distribution,in,r,s,t}^{flow} + v_{battery,r,s,t}^{charge} &= v_{electricity_transport,out,r,s,t}^{flow} + v_{nuclear_unit,r,s,t}^{flow} + v_{gas_unit,out,r,s,t}^{flow} + \\
v_{electricity_transport,in,r,s,t}^{flow} &= v_{wind_unit,r,s,t}^{flow} + v_{electricity_import,out,r,s,t}^{flow}
\end{aligned}$$

$$\begin{aligned}
v_{gas_distribution,out,r,s,t}^{flow} &= p_{gas_distribution}^{efficiency} \cdot v_{gas_distribution,in,r,s,t}^{flow} \\
v_{electricity_distribution,out,r,s,t}^{flow} &= p_{electricity_distribution}^{efficiency} \cdot v_{electricity_distribution,in,t}^{flow} \\
v_{electricity_transport,out,r,s,t}^{flow} &= p_{electricity_transport}^{efficiency} \cdot v_{electricity_transport,in,r,s,t}^{flow}
\end{aligned}$$

$$\sum_{t \in t_{gas}} v_{gas_unit,out,r,s,t}^{flow} \cdot 1h = p_{gas_unit}^{efficiency} \cdot v_{gas_unit,in,r,s,t_{gas}}^{flow} \cdot \Delta_{t_{gas}}$$

$$\begin{aligned}
v_{nuclear_unit,r,s,t}^{on} &= v_{nuclear_unit,r,s,t-1}^{on} + v_{nuclear_unit,r,s,t}^{on_up} - v_{nuclear_unit,r,s,t}^{on_down} \\
v_{nuclear_unit,r,s,t}^{on_up} &\leq 1 - v_{nuclear_unit,r,s,t}^{on} - \sum_{t'=1}^{MDT-1} v_{nuclear_unit,r,s,t-t'}^{on_down} \\
v_{nuclear_unit,r,s,t}^{on_down} &\leq v_{nuclear_unit,r,s,t}^{on} - \sum_{t'=1}^{MUT-1} v_{nuclear_unit,r,s,t-t'}^{on_up}
\end{aligned}$$

$$\begin{aligned}
v_{battery,r,s,t}^{state,intra} &= p_{battery}^{efficiency} \cdot v_{battery,r,s,t-1}^{state,intra} + p_{battery}^{efficiency,charge} \cdot v_{battery,r,s,t}^{charge} - v_{battery,r,s,t}^{discharge} \\
v_{battery,s,d}^{state,inter} &= v_{battery,s,d-1}^{state,inter} + \sum_r W_{r,d} \cdot (v_{battery,r,s,0}^{state,intra} - v_{battery,r,s,T}^{state,intra}) \\
v_{storage,s,0}^{state,inter} &= v_{storage,s,D}^{state,inter}
\end{aligned}$$

bound to:

$$\begin{aligned}
0 &\leq v_{gas_distribution,out,r,s,t}^{flow} && \leq p_{gas_distribution}^{capacity} \\
0 &\leq v_{electricity_distribution,out,r,s,t}^{flow} && \leq p_{electricity_distribution}^{capacity} \cdot p_{electricity_distribution}^{existing_units} \\
0 &\leq v_{electricity_transport,out,r,s,t}^{flow} && \leq p_{electricity_transport}^{capacity} \cdot p_{electricity_transport}^{existing_units} \\
\\
0 &\leq v_{gas_import,r,s,t}^{flow} && \leq p_{gas_import}^{capacity} \\
0 &\leq v_{electricity_import,r,s,t}^{flow} && \leq p_{electricity_import}^{capacity} \\
\\
p_{nuclear_unit}^{capacity_min} \cdot p_{nuclear_unit}^{existing_units} \cdot v_{nuclear_unit,r,st}^{on} &\leq v_{nuclear_unit,r,s,t}^{flow} && \leq p_{nuclear_unit}^{capacity} \cdot p_{nuclear_unit}^{existing_units} \cdot v_{nuclear_unit,r,s,t}^{on} \\
0 &\leq v_{gas_unit,in,r,s,t}^{flow} && \leq \frac{1}{p_{gas_unit}^{efficiency}} \cdot p_{gas_unit}^{capacity} \\
0 &\leq v_{gas_unit,out,r,s,t}^{flow} && \leq p_{gas_unit}^{capacity} \\
\\
0 &\leq v_{nuclear_unit,r,s,t}^{on} \\
0 &\leq v_{nuclear_unit,r,s,t}^{on_up} \\
0 &\leq v_{nuclear_unit,r,s,t}^{on_down} \\
\\
0 &\leq v_{solar_unit,r,s,t}^{flow} && \leq p_{solar_unit,r,s,t}^{flow_profile} \cdot p_{solar_unit}^{capacity} \\
0 &\leq v_{wind_unit,r,s,t}^{flow} && \leq p_{wind_unit,r,s,t}^{flow_profile} \cdot p_{wind_unit}^{capacity} \\
\\
0 &\leq v_{battery,r,s,t}^{charge} && \leq p_{battery}^{capacity_charge} \\
0 &\leq v_{battery,r,s,t}^{discharge} && \leq p_{storage}^{capacity_discharge} \\
0 &\leq v_{battery,r,s,t}^{state,intra} && \leq p_{storage}^{capacity} \\
0 &\leq v_{battery,s,d}^{state,inter} && \leq p_{storage}^{capacity} \\
\\
p_{gas_unit}^{existing_units} &\leq v_{gas_unit}^{investment} \\
p_{nuclear_unit}^{existing_units} &\leq v_{nuclear_unit}^{investment} \\
p_{solar_unit}^{existing_units} &\leq v_{solar_unit}^{investment} \\
p_{wind_unit}^{existing_units} &\leq v_{wind_unit}^{investment} \\
p_{battery}^{existing_units} &\leq v_{battery}^{investment}
\end{aligned}$$

Appendix A

ines spec

The intended way to inspect the ines spec is through Spine Toolbox. For reference, a static table is provided below.

<i>entity class name</i>	<i>parameter name</i>	<i>valid types</i>
constraint	constant	float
constraint	sense	
link	availability	float 1d_map time_series
link	availability_forecasts	1d_map 2d_map
link	capacity	float
link	constraint_link_count_coefficient	1d_map
link	efficiency	float 1d_map time_series
link	efficiency_forecasts	1d_map 2d_map
link	fixed_co2_emissions	float 1d_map
link	fixed_cost	float 1d_map
link	flow_max_cumulative	float 1d_map
link	flow_min_cumulative	float 1d_map
link	interest_rate	float 1d_map
link	investment_co2_emissions	float 1d_map
link	investment_cost	float 1d_map
link	investment_method	
link	investment_uses_integer	
link	is_non_synchronous	
link	lifetime	float 1d_map
link	links_existing	float 1d_map
link	links_fix_cumulative	float 1d_map
link	links_invest_fix_period	1d_map
link	links_invest_max_period	1d_map
link	links_invest_min_period	1d_map
link	links_max_cumulative	float 1d_map
link	links_min_cumulative	float 1d_map
link	links_retire_period	1d_map
link	operational_cost	float 1d_map time_series
link	operational_cost_forecasts	1d_map 2d_map
link	retirement_method	
link	salvage_value	float 1d_map
link	transfer_method	
node	co2_content	float 1d_map
node	commodity_limit	float 1d_map
node	commodity_price	float 1d_map time_series
node	commodity_price_forecasts	1d_map 2d_map
node	commodity_price_ladder	1d_map
node	constraint_storage_count_coefficient	1d_map
node	constraint_storage_state_coefficient	1d_map
node	flow_annual	float 1d_map
node	flow_profile	float 1d_map time_series
node	flow_profile_forecasts	1d_map 2d_map
node	flow_scaling_method	
node	node_type	
node	penalty_downward	float 1d_map
node	penalty_upward	float 1d_map
node	spill_lower_limit	float 1d_map time_series
node	spill_upper_limit	float 1d_map time_series
node	storage_capacity	float
node	storage_cost_of_stored_energy	float 1d_map
node	storage_fixed_co2_emissions	float 1d_map
node	storage_fixed_cost	float 1d_map
node	storage_interest_rate	float 1d_map
node	storage_investment_co2_emissions	float 1d_map
node	storage_investment_cost	float 1d_map
node	storage_investment_method	
node	storage_investment_uses_integer	

value

s

link_inve

bo

bo

link_retire

link_tran

flow_scal

nod

storage_inve

bo

Appendix B

Formulation ines [work in progress]

From the certification process, it is possible to piece together the overall intended formulation of the ines spec. As the certification process is currently only a proof of concept, this formulation is not yet complete and as such a work in progress.

The ines spec is disentangled here in the typical formulation of a linear program: indices, parameters, variables, objective and constraints. The formulation parameters and variables are formatted as $p_{indices}^{name}$ and $v_{indices}^{name}$ respectively. The names correspond directly to the input parameters in the ines spec, unless otherwise specified in the preprocessing section. The indices are always the most complex possible datastructure making the equations valid for all possible input values.

B.1 Indices

set groups of units, connections, etc.

set_{unit} set that a unit belongs to

set_{connection} set that a connection belongs to

set_{node} set that a node belongs to

period investment periods in the system

time time steps (always continuous within period or rolling horizon, weight is used for representative days)

time_{set} time steps in set

period_{time} period corresponding to time

unit units in the system

unit__to_node unit to nodes in the system

unit_{unit__to_node} unit in the unit__to_node entity

node__to_unit unit to nodes in the system

unit__node__to_unit unit in the *unit__to_node* entity

B.2 Parameters

The parameters are formatted as $p_{sets}^{parameter_name}$ with the parameter name from the table below and the set based on the maximum of the valid types.

node__to_unit has the same parameters as *unit__to_node* so that is not repeated here

$p_{unit}^{interest_rate}$ discount rate related to a unit

$p_{unit,period,time}^{startup_cost}$ startup cost

$p_{unit,period,time}^{shutdown_cost}$ shutdown cost

$p_{unit,period,time}^{online_cost}$ online cost

$p_{unit_to_node,period}^{investment_cost}$ investment cost of a unit flow in a certain period

$p_{unit_to_node,period}^{fixed_cost}$ FOM cost of a unit flow

$p_{un,p,t}^{other_operational_cost}$ VOM cost of a unit flow

$p_{unit_to_node,period}^{ramp_cost}$ the ramp cost is specifically described to be the cost of ramping up (not ramping down)

B.3 Variables

According to the ines-output spec, for a *node__to_unit*, there is a capacity, a flow per period and a flow per time step.

node__to_unit is the same as *unit__to_node* so it is not repeated here.

$v_{un,p,t}^{commit}$ committed units

$v_{un,p,t}^{commit_up}$ units that started up

$v_{un,p,t}^{commit_down}$ units that shut down

$v_{unit_to_node,period}^{investment}$ additional capacity of installed *unit__to_node* in a period

$v_{unit_to_node,period}^{capacity}$ capacity of *unit__to_node* in each period

$v_{unit_to_node,period,time}^{flow}$ flow of *unit__to_node* in each time in each period

$v_{unit_to_node,period,time}^{rampup}$ ramp up of *unit__to_node* in each time in each period

B.4 Objective

min

$$\begin{aligned}
& + \sum_{u \in \text{unit}} \sum_{p \in \text{period}} p_{u,p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p} \Delta_t \cdot p_{un,p}^{\text{online_cost}} \cdot v_{un,p,t}^{\text{commit}} \\
& + \sum_{u \in \text{unit}} \sum_{p \in \text{period}} p_{u,p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p \notin 1} p_{un,p}^{\text{startup_cost}} \cdot v_{un,p,t}^{\text{commit_up}} \\
& + \sum_{u \in \text{unit}} \sum_{p \in \text{period}} p_{u,p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p \notin 1} p_{un,p}^{\text{shutdown_cost}} \cdot v_{un,p,t}^{\text{commit_down}} \\
& + \sum_{un \in \text{unit_to_node}} \sum_{p \in \text{period}} p_{unit[un],p}^{\text{economic_investment}} \cdot p_{un,p}^{\text{investment_cost}} \cdot v_{un,p}^{\text{investment}} \\
& + \sum_{un \in \text{unit_to_node}} \sum_{p \in \text{period}} p_{unit[un],p}^{\text{economic_fixed}} \cdot p_{un,p}^{\text{fixed_cost}} \cdot v_{un,p}^{\text{capacity}} \\
& + \sum_{un \in \text{unit_to_node}} \sum_{p \in \text{period}} p_{unit[un],p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p} \Delta_t \cdot p_{un,p,t}^{\text{other_operational_cost}} \cdot v_{un,p,t}^{\text{flow}} \\
& + \sum_{un \in \text{unit_to_node}} \sum_{p \in \text{period}} p_{unit[un],p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p \notin 1} p_{un,p}^{\text{ramp_cost}} \cdot v_{un,p,t}^{\text{ramp_up}} \\
& + \sum_{nu \in \text{node_to_unit}} \sum_{p \in \text{period}} p_{unit[nu],p}^{\text{economic_investment}} \cdot p_{nu,p}^{\text{investment_cost}} \cdot v_{nu,p}^{\text{investment}} \\
& + \sum_{nu \in \text{node_to_unit}} \sum_{p \in \text{period}} p_{unit[nu],p}^{\text{economic_fixed}} \cdot p_{nu,p}^{\text{fixed_cost}} \cdot v_{nu,p}^{\text{capacity}} \\
& + \sum_{nu \in \text{node_to_unit}} \sum_{p \in \text{period}} p_{unit[nu],p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p} \Delta_t \cdot p_{nu,p,t}^{\text{other_operational_cost}} \cdot v_{nu,p,t}^{\text{flow}} \\
& + \sum_{nu \in \text{node_to_unit}} \sum_{p \in \text{period}} p_{unit[nu],p}^{\text{economic_other}} \cdot \sum_{t \in \text{time}_p \notin 1} p_{nu,p,t}^{\text{ramp_cost}} \cdot v_{nu,p,t}^{\text{ramp_up}}
\end{aligned}$$

B.5 Reserves

$\forall n \in \text{node_reserve} :$

$$\begin{aligned}
& \sum_{u \in n} r_{u,t}^+ + \sum_{u \in su \cup n} r_{u,t}^{c,+} + r_{u,t}^{d,+} + \sum_{u \in dlu \cup n} r_{u,t}^{l,+} + \sum_{u \in du \cup n} r_{u,t}^{ll} \\
& - \sum_{u \in du \cup n} F^{\text{dres}} \cdot P_{u,Y_t}^{\text{peak}} - \sum_{u \in ru \cup n} F^{\text{res}} \cdot c_{u,Y_t} = 0 \\
& \sum_{u \in n} r_{u,t}^- + \sum_{u \in su \cup n} r_{u,t}^{c,-} + r_{u,t}^{d,-} + \sum_{u \in dlu \cup n} r_{u,t}^{l,-} \\
& - \sum_{u \in du \cup n} F^{\text{dres}} \cdot P_{u,Y_t}^{\text{peak}} - \sum_{u \in ru \cup n} F^{\text{res}} \cdot c_{u,Y_t} = 0
\end{aligned}$$

B.6 Unit Constraints

$$\begin{aligned}v_{un,p,t}^{ramp-up} &> 0 \\v_{un,p,t}^{ramp-up} &> v_{un,p,t}^{flow} - v_{un,p,t-1}^{flow} \\v_{un,p,t}^{ramp-down} &> 0 \\v_{un,p,t}^{ramp-down} &> v_{un,p,t-1}^{flow} - v_{un,p,t}^{flow}\end{aligned}$$

Appendix C

Extended Certification process

The certification setups from Chapter 2 are selected for a proof of concept. The full certification process consists of:

equation tests are a systematic group of tests to test each equation of energy system models individually, starting from a simple energy balance on one node with 1 demand unit and on supply unit

method tests are tests that cover multiple equations to show off a certain feature (various examples can be found in Chapter 2)

system tests are more comprehensible, similar to IEEE networks (an example can be found in the final setup of Chapter 2)

For the equation tests, inspiration can be taken from the different formulation in the proof of concept of the certification process. For the method tests, table C.1 is an indicative expansion of the certification process. The system tests are a point of discussion for the consortium of the ines spec.

name	
network = copper plate	
network = radial	
network = meshed	
resolution = fixed	
resolution = variable	
resolution = flexible	
stochastic	
stochastic = flexible	
CO2 cap	
reserves	
cogeneration	
ramping	
flow = fixed	
flow = limited	
storage = intra	
storage = inter	
part load efficiency	
MUT	Meshed network, fixed resolution, expensive storage and cheap unit. By s
MDT	
economic representation	
investment = fixed	
investment = cumulative	
investment = multi year	
retirement = fixed	
retirement = economic	
representative periods	
blended representative periods	
rolling horizon	
benders decomposition	
mini Europe	

Table C.1: Expansion of the certification process.